Year 13 Maths - Further trigonometry

Similar questions - Expressing a sin x + b cos x as a single function

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Express $8\sin x + 6\cos x$ in the form $R\sin(x+\alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$, giving α to 4 decimal places. **[3]**

(b) State the greatest value of $8\sin x + 6\cos x$ and the smallest positive value of x at which it occurs. **[2]**

(c) Solve $8\sin x + 6\cos x = 5$ for $0 \leqslant x \leqslant 2\pi$, giving your answers to 3 significant figures. **[4]**

2 The displacement x metres of a particle at time t seconds is modelled by $x = 3\cos 2t - 4\sin 2t$ for $t \geqslant 0$.

(a) Express x in the form $R\cos(2t+\alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$. **[3]**

(b) Write down the greatest displacement and find the first time at which it occurs. **[3]**

(c) Find the times in the first 3 seconds at which the displacement is 2.5 m. **[4]**

Question 1

(a) $R = \sqrt{8^2+6^2} = 10$, $\tan\alpha = \frac{6}{8}$ so $\alpha = 0.6435$. $8\sin x + 6\cos x = 10\sin(x+0.6435)$

(b) Greatest value 10 when $x+\alpha = \frac{\pi}{2}$, so $x = \frac{\pi}{2} - 0.6435 = 0.927$

(c) $\sin(x+0.6435) = 0.5$, so $x+0.6435 = \frac{\pi}{6},\ \frac{5\pi}{6},\ \frac{13\pi}{6}$. The first gives a negative x (rejected). $x = \frac{5\pi}{6} - 0.6435 = 1.97$ and $x = \frac{13\pi}{6} - 0.6435 = 6.16$

Question 2

(a) $R\cos\alpha = 3$, $R\sin\alpha = 4$, so $R = 5$ and $\tan\alpha = \frac{4}{3}$, $\alpha = 0.9273$. $x = 5\cos(2t+0.9273)$

(b) Greatest displacement 5 m, when $2t+\alpha = 2\pi$ (the first positive time), so $t = \frac{2\pi - 0.9273}{2} = 2.68$ s

(c) $\cos(2t+0.9273) = 0.5$, so $2t+0.9273 = \frac{\pi}{3},\ \frac{5\pi}{3},\ \frac{7\pi}{3}$. Then $t = 0.0600$ s, 2.15 s and 3.20 s. The first two are within 3 seconds: $t = 0.0600$ and $t = 2.15$